Diseases of the Pancreas

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

Type 1 diabetes, type 2 diabetes, diabetes labs, and pancreatitis

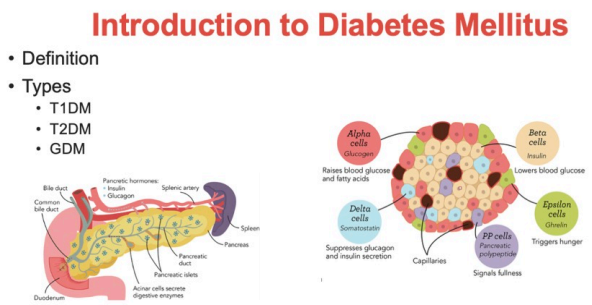
Lecture roadmap: The pancreas's two jobs (exocrine and endocrine) → What diabetes is → Type 1 diabetes and DKA → Diagnostic labs → Type 2 diabetes and HHS → Understanding the labs (FPG, A1C, OGTT, random, antibodies, C-peptide) → Why type 1 shows up fast and type 2 slowly → Pancreatitis (acute and chronic)

How the instructor wants you to think: Ask **which pancreatic function is affected** and how that explains the findings. Focus on the **differences between type 1 and type 2** and connect them to manifestations and labs. **Gestational diabetes appears in your ATI Engage module but is NOT covered in this course**; it comes later in maternal–newborn. Management is not tested beyond the basics noted below.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further or a transcript slip cleaned up; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

The Pancreas: Two Jobs

Exocrine (digestion)	Endocrine (glucose regulation)
Acinar cells make **digestive enzymes** that travel through **ducts** into the small intestine (covered in the GI module). Disrupted in **pancreatitis**.	**Islet cells** release hormones **directly into the blood**: **beta cells** → **insulin** (lowers glucose); **alpha cells** → **glucagon** (raises glucose). Disrupted in **diabetes**.



From the lecture slides: the pancreas and the cell types in a pancreatic islet.

Diabetes Mellitus: The Basics

Persistently elevated blood glucose because the body **doesn't make enough insulin**, **doesn't respond effectively to insulin**, or **both**.

What insulin normally does: Moves glucose **into skeletal muscle and adipose tissue**, and tells the **liver to store glucose** and **reduce its release** into the blood.
Type 1: the cells that **make** insulin are **destroyed**. **Type 2:** tissues **respond less** to insulin that's present (**insulin resistance**), and eventually the pancreas can't make enough to keep up.

Type 1 Diabetes (T1DM)

Etiology	An **autoimmune** response destroys the pancreatic **beta cells**, so insulin production fails. Often linked with childhood, but it **can develop at any age**.
Pathophysiology	No insulin → muscle and fat **can't take up glucose**, and the **liver keeps releasing glucose** → blood glucose rises even though the body **can't use or store energy** properly.
Manifestations	• **Polyuria**: filtered glucose exceeds what the kidneys can reabsorb → glucose stays in the urine and **holds water in the tubules** = **osmotic diuresis** • **Polydipsia**: fluid loss → intense thirst • **Increased hunger**, **fatigue**, **unintentional weight loss**: cells can't use glucose, so the body **breaks down fat and protein** for energy • Slide images also show blurry vision, slow-healing wounds, and dry itchy skin
DKA	**Fat breakdown** gives the liver material to make **ketones** → ketones accumulate → **metabolic acidosis**. This is the link between **severe insulin deficiency** and **diabetic ketoacidosis (DKA)**. Slide image findings: **vomiting**, **abdominal pain**, **fruity-smelling breath**, **Kussmaul respirations**, polydipsia, polyuria.
Diagnosis	Glucose and A1C thresholds (see the labs table); **pancreatic antibodies** support an autoimmune process; **C-peptide is typically low** as beta cells are lost.

Impact on health	**Persistent hyperglycemia damages blood vessels and nerves**: **kidney** damage (glomeruli; connects to the renal module), **retinal damage**, **neuropathy**, **cardiovascular disease**.
Management	**Insulin replacement is required to stay alive**, because the beta cells no longer make insulin. Nutrition, activity, and glucose monitoring help but cannot replace insulin.

Type 2 Diabetes (T2DM)

Pathophysiology	Insulin **is present**; the beta cells are not destroyed at first. **Tissues don't respond to it normally**, which is **insulin resistance**. 1. **Early:** the pancreas **compensates** by releasing **more insulin**, so glucose can stay in range for a while. 2. **Over time: beta cell function declines**. When insulin output can't meet the body's needs, glucose rises.
Etiology / risk factors	**Genetic susceptibility** (a big factor), **aging**, **adiposity** (more body fat), **physical activity**. **Instructor emphasis: check your biases.** Type 2 diabetes is **not a moral failing**. Genetic metabolic dysfunction can itself **cause** increased adiposity (a chicken-and-egg situation), and very active people can still develop T2DM. Don't reduce it to someone's choices.
Manifestations	Develops **gradually**; a person may have it for a long time before noticing. **Fatigue**, **blurry vision**, **recurrent infections**, **slow wound healing**. Polyuria and polydipsia can occur (same osmotic diuresis) but are **more of a type 1 hallmark**.
HHS	**Hyperosmolar hyperglycemic state**: **very high glucose** → profound urinary water loss → **severe dehydration** → blood becomes **concentrated** → affects **neurologic function**. Usually there's **enough insulin to limit ketone production**, though not enough to control glucose, so **DKA is much less likely** in type 2.
Diagnosis	**Same glucose and A1C thresholds as type 1.** The distinction comes from the underlying process and clinical evaluation.
Impact on health	Same long-term **vascular and nerve injury**: neuropathy, nephropathy, retinopathy, heart attack and heart failure, stroke and cognitive decline, poor circulation. **Gradual onset ≠ harmless**: complications may already be developing before diagnosis.

Pharm connection	Drugs target different parts of the problem: **reduce liver glucose production**, **improve insulin action**, or affect **insulin release** or **glucose elimination**. **Some people with type 2 need insulin** as their own production declines; that reflects disease progression.

DKA vs. HHS (instructor emphasis): **Associate DKA with type 1 and HHS with type 2**, but **never say "always" or "never"**: people with type 2 can develop DKA.

	DKA (typical in type 1)	HHS (typical in type 2)
Insulin	Severe **deficiency**	Some insulin present: enough to limit ketones, not enough to control glucose
Mechanism	Fat breakdown → **ketones** → **metabolic acidosis**	**Very high glucose** → osmotic diuresis → **profound dehydration**, concentrated blood
Key effects	Kussmaul respirations, fruity breath, vomiting, abdominal pain	**Neurologic changes** from the hyperosmolar state

Understanding the Diabetes Labs

Thresholds are the **same for type 1 and type 2**.

Test	What it tells us	Diabetes threshold
Fasting plasma glucose	Glucose after **≥ 8 hours** with no caloric intake: how the body regulates glucose **without a recent meal**	**≥ 126 mg/dL**
A1C	Glucose attaches to **hemoglobin** in RBCs. The percentage with glucose attached estimates the **average glucose over about 3 months**. **No fasting needed.**	**≥ 6.5%** Normal < 5.7% Prediabetes 5.7–6.4%

Test	What it tells us	Diabetes threshold
2-hour oral glucose tolerance test	Drink **75 g glucose**, measure 2 hours later. Normally insulin brings glucose back down (≤ 199).	**≥ 200 mg/dL**
Random plasma glucose	Any time, regardless of meals	**≥ 200 mg/dL** *with classic symptoms* (thirst, frequent urination, weight loss) or a hyperglycemic crisis

- Without clear symptomatic hyperglycemia, an abnormal result generally needs **confirmation** with a second test (details are beyond this course).
- A **glucose** level shows **right now**; the **A1C** shows the **pattern over time**.
- Labs may report slightly different ranges; the A1C is the most standardized.

Which Type? Antibodies and C-Peptide

The tests above show **whether** someone has diabetes, **not which type**.

- **Pancreatic antibodies** = evidence of an **autoimmune** attack on insulin-producing cells → supports **type 1**.
- **C-peptide** is released **alongside the body's own insulin**. **Injected insulin contains no C-peptide**, so C-peptide measures the person's **own** insulin production, even if they're on insulin.
  - **Type 1:** beta cells lost → **C-peptide low**
  - **Type 2: normal or even high** early (the pancreas is overworking against resistance), then **declines** as the disease progresses

Pause and Reflect: Why can type 1 become noticeable quickly while type 2 goes unrecognized for years?
Type 1: a **substantial loss of insulin production** causes a **rapid** change in glucose regulation, so marked **thirst, frequent urination, and weight loss** appear quickly.
Type 2: no beta cell destruction at first, so the pancreas can **compensate** for insulin resistance and **mask** the onset. Hyperglycemia develops only as compensation fails, so symptoms appear **gradually**, and they are **nonspecific** (fatigue, blurry vision) and easy to overlook.
Keep it flexible: type 2 can present with **severe** hyperglycemia and sometimes DKA. These are **typical patterns, not absolute rules**.

Type 1 vs. Type 2 at a Glance

Feature	Type 1	Type 2
Core problem	**Autoimmune destruction of beta cells** → no insulin	**Insulin resistance** → later, declining beta cell function
Onset	**Rapid**; any age (often childhood)	**Gradual**; may go unrecognized for years
Key symptoms	Polyuria, polydipsia, hunger, **weight loss**, fatigue	Fatigue, blurry vision, recurrent infections, slow healing
Acute crisis	**DKA** (ketones → metabolic acidosis)	**HHS** (extreme glucose → dehydration, neuro changes)
Antibodies	Present	Absent
C-peptide	**Low**	Normal or high early → declines later
Treatment basis	**Insulin required**	Various drugs; some eventually need insulin

Pancreatitis

A shift from glucose regulation to **injury of pancreatic tissue**.

Normal protection: Many **protein-digesting enzymes leave the pancreas in an inactive form** and are activated later in the intestine. The pancreas is made of protein, so this keeps it from **digesting itself**.

Pathophysiology	Pancreatic injury and **premature enzyme activation** → enzymes digest the pancreas's own tissue (**autodigestion**) → **inflammation and tissue damage**.
Etiology	• **Gallstones**: block pancreatic drainage near the **shared opening** into the small intestine (connects to the gallbladder lecture) • **Alcohol** (common) • **Very high triglycerides**, certain **medications**, **pancreatic injury** • Not all pancreatitis starts with a blocked duct.

Acute pancreatitis	• **Severe epigastric pain**, often **radiating to the back** • **Nausea and vomiting** • Inflammation + **increased capillary permeability** → fluid moves **out of the circulation** into surrounding tissue. In severe cases this **reduces circulating volume and impairs perfusion**.
Chronic pancreatitis	**Lasting damage and fibrosis**. Functional tissue is lost: • **Fewer digestive enzymes** → **steatorrhea** (bulky, greasy stools from poor fat digestion), **maldigestion**, **weight loss**, **nutritional deficiencies** • Damage to **insulin-producing cells** → impaired glucose regulation → **diabetes**
Diagnosis	• **Lipase** is especially useful: pancreatic injury raises its blood level. • **Imaging** shows pancreatic changes, complications, or the cause, but **isn't always needed** for acute pancreatitis when the presentation and enzymes already fit. • **A higher lipase does NOT mean more severe disease.** Severity depends on effects on **circulation and other organs**. • **Chronic:** enzymes can be **normal despite major damage**, because so much tissue is lost that large elevations no longer occur.

Instructor wrap-up: **Type 1** = loss of insulin production → hyperglycemia and ketone risk. **Type 2** = insulin resistance, then declining beta cell function → impaired glucose regulation. **Pancreatitis** = injury that disrupts **digestion** and, over time, **insulin production**. Always ask which function is disrupted and how that explains the findings.

Self-Check ADDED

1. Why does high blood glucose cause polyuria?
   Glucose exceeds the kidneys' reabsorption capacity, stays in the tubules, and pulls water with it (osmotic diuresis).
2. Why does a person with new type 1 diabetes lose weight despite eating more?
   Without insulin, cells can't use glucose, so the body breaks down fat and protein for energy.
3. How does insulin deficiency lead to metabolic acidosis?
   Fat breakdown supplies the liver with material to make ketones, which are acids.

4. Why is DKA less likely in type 2?
   There is usually enough insulin to limit ketone production, though not enough to control glucose.
5. Which test reflects average glucose over about 3 months, and does it require fasting?
   A1C. No fasting needed.
6. A client has an A1C of 6.0%. How is this classified?
   Prediabetes (5.7–6.4%).
7. A random glucose of 240 mg/dL in a client with thirst, frequent urination, and weight loss: is this enough for diagnosis?
   Yes. Random ≥200 with classic symptoms establishes the diagnosis.
8. Why can C-peptide assess insulin production in someone taking insulin?
   Injected insulin has no C-peptide, so C-peptide reflects only the person's own production.
9. Why is C-peptide often normal or high early in type 2?
   The pancreas is overproducing insulin to overcome resistance.
10. What normally prevents the pancreas from digesting itself?
    Enzymes leave the pancreas in an inactive form and are activated later in the intestine.
11. Why can acute pancreatitis cause hypovolemia?
    Inflammation increases capillary permeability, so fluid shifts out of the circulation into the tissues.
12. A client with chronic pancreatitis has greasy, bulky stools and a normal lipase. Explain both.
    Steatorrhea from too few digestive enzymes. Lipase can be normal because too little functional tissue remains to release large amounts.
13. Does a very high lipase mean severe pancreatitis?
    No. Severity depends on effects on circulation and other organs.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Pancreas, NURA 340, ATI Engage deck) and the lecture transcript. Most diabetes slides are images, so much of the content comes from the narration; the labs table and the DKA vs. HHS details come from slide images.

- CLARIFIED **"B cells" = beta cells.** The captions write "B cells"; the lecturer means pancreatic **beta** cells (insulin-producing), not B lymphocytes.
- CLARIFIED **Chronic pancreatitis and lipase.** The lecturer first says the person "may not have lost enough functional tissue," then corrects herself: they may have lost **so much** tissue that large enzyme elevations no longer occur. The guide uses the corrected version.
- CLARIFIED **Secondary diabetes.** In the pituitary/adrenal lecture, the instructor notes that diabetes caused by cortisol excess (Cushing's) is called secondary diabetes. Chronic pancreatitis causing diabetes is a similar idea.
- CLARIFIED **DKA manifestations** (Kussmaul respirations, fruity breath, vomiting, abdominal pain) come from slide images rather than the narration, and are included because they match the mechanism the lecturer explains.
- ADDED **DKA vs. HHS table** and **Type 1 vs. Type 2 at a Glance** reorganize lecture content side by side; the **Self-Check questions** are my own.
- **Gestational diabetes is intentionally left out**, per the instructor. It appears in ATI Engage but is not covered in this course.
- Insulin delivery methods and continuous glucose monitors appear in slide images; they are management details not tested in this course.